MTH2010 Algebra

Problem Sheet 1

During tutorials in Week 2, questions 4 and 6 will be discussed.

1. **(a)** Find the order of each element of the group $(\mathbf{Z}_8, +)$. Do the same for the elements of $(\mathbf{Z}_{12}, +)$.
(b) What are the possible orders of the elements of the dihedral group D_{2n}? Show that each element of D_{2n} which is not a power of r has order 2. Explain why if n is odd, and if $d > 2$ is a divisor of n, then D_{2n} has no cyclic subgroup of cardinality $2d$.

2. Let G be a group and let x be an element of G.
(a) Show that $\text{ord}(x) = \text{ord}(x^{-1})$.
(b) Let $y \in G$. Show that $\text{ord}(x) = \text{ord}(yxy^{-1})$. Deduce that $\text{ord}(ab) = \text{ord}(ba)$ for all $a, b \in G$.

3. **(a)** **(i)** Find all cyclic subgroups of D_8. Find a subgroup of D_8, distinct from D_8, which is not cyclic.
(ii) Find a non cyclic subgroup of D_{16} of cardinality 4. Find a non cyclic subgroup of D_{16} of cardinality 8.
(b) Given positive integers n and $d > 1$, with d divides n, find a cyclic subgroup of D_{2n} of cardinality d. Find a non cyclic subgroup of D_{2n} of cardinality $2d$.

4. **(a)** Let $G = \langle x \rangle$ be a cyclic group generated by an element $x \in G$. Prove that every subgroup of G is cyclic.
(b) If $G = \langle x \rangle$ is a finite cyclic group of cardinality n, and if d is a positive divisor of n, show that G has a **unique** subgroup of cardinality d.

5. Let A and B be groups and let $A \times B$ be their product.
(a) Prove that $A \times B$ is abelian if and only if both A and B are abelian.
If $A \times B$ is cyclic show that both A and B are cyclic. Is the converse true? Justify your answer.
(b) Let $C \subset A$ and $D \subset B$ be subgroups. Prove that the set

$$C \times D \stackrel{\text{def}}{=} \{(c, d) \in A \times B : c \in C,\ d \in D\}$$

is a subgroup of $A \times B$. Find a subgroup of $\mathbb{Z} \times \mathbb{Z}$ which doesn't occur in this way, i.e. find a subgroup of $\mathbb{Z} \times \mathbb{Z}$ which is not the product of 2 subgroups of $\mathbb{Z}$.
(c) Let $a \in A$, $b \in B$, with $\text{ord}(a) = n < \infty$ and $\text{ord}(b) = m < \infty$. Show that the order of $(a, b) \in A \times B$ is the least common multiple $\text{lcm}(n, m)$ of the orders of a and b (observe that the elements (a, e_B) and (e_A, b) commute and use Q(7) below). What is the order of (a, b) if $\text{ord}(a) = \infty$ or $\text{ord}(b) = \infty$? Justify your answer.

6. Let $\phi : G \to H$ be a group homomorphism and let $x \in G$.
(a) Prove that $\phi(x^n) = \phi(x)^n$ for all $n \in \mathbb{Z}$ (consider first the cases $n \geq 0$, $n = -1$, and then the case $n < -1$).
(b) Assume that $\text{ord}(x)$ finite. Prove that $\text{ord}(\phi(x))$ is finite and divides $\text{ord}(x)$. Further prove that $\text{ord}(\phi(x)) = \text{ord}(x)$ if ϕ is an isomorphism. Give an example of a homomorphism ϕ as above where:

(i) $\mathrm{ord}(x) = \infty$, $\mathrm{ord}(\phi(x))$ finite.
(ii) $\mathrm{ord}(x) = \mathrm{ord}(\phi(x)) = \infty$.

7. If a and b are elements of a group G such that $ab = ba$, and if $n \geq 1$, show that $(ab)^n = a^n b^n$.